

# *M.Sc. Rocío Nayeli Avendaño Villeda*

## *Environmental Consultant & Marine Spatial Data Analyst*

## CONTACT


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La Paz, Baja California Sur, Mexico.


## LANGUAGES

**Spanish: Native 100%**

**English: Intermediate**

## INTERPERSONAL SKILLS

- ✓ Cross-functional collaboration
- ✓ Problem solving
- ✓ Proactive
- ✓ Disciplined
- ✓ Punctual
- ✓ Flexible
- ✓ Responsible

## PROFILE

Marine Biologist and Environmental Data Analyst with expertise in coastal ecosystem management, spatial data modeling (GIS, R, Python), and environmental sustainability standards (MSC). Proven track record in conducting environmental baseline studies, assessing human-ecosystem interactions (benthic impact, protected species bycatch), and implementing mitigation protocols. Experienced in multi-stakeholder training and field data collection for coastal and marine resource management.

## CORE COMPETENCIES & TECHNICAL SKILLS

**Environmental Impact & Baseline Assessment:** Spatial modeling of benthic substrates, fauna interaction monitoring, vulnerability mapping, and environmental risk evaluation. GIS & Spatial Analysis QGIS, Spatial Databases (PostgreSQL), Remote Sensing (SST, Chlorophyll-*a*, Altimetry via NASA/Copernicus).

**Data Science & Programming:** R (`terra`, `sf`, `dplyr`, time series, multivariate stats), Python (`Geopandas`, `Numpy`, `Matplotlib`), SQL, Advanced Excel.

**Regulatory & Sustainability Frameworks:** Marine Stewardship Council (MSC) Standards, Mexican Environmental Regulations (LGEEPA, NOM-059, NOM-003, NOM-022), Adaptive Fisheries & Resource Management.

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## PROFESSIONAL EXPERIENCE

**Marine Spatial Data Analyst & Sustainability Specialist Alianza por la Sostenibilidad de la Sardina (ALISSA A.C.) Jan 2024 – Present**

- Performed spatial modeling to evaluate the physical interaction between commercial fishing operations and benthic habitats, identifying vulnerable seabed zones to support sustainable management plans.
- Analyzed long-term oceanographic datasets (MODIS, Copernicus altimetry, upwelling indices) and catch statistics to establish baseline environmental conditions and assess climate variability impacts.
- Modeled spatiotemporal dynamics of incidental bycatch involving protected seabird species, defining high-risk geographic zones and seasonal hotspots to design targeted mitigation strategies.
- Conducted capacity-building workshops and training programs for vessel captains, fleet managers, and onboard observers on biological sampling protocols, bycatch reduction, and MSC compliance standards.
- Prepared comprehensive technical and climatic assessment reports to support industry decision-making under precautionary environmental frameworks.

## DATA ANALYSIS PROJECTS

- Processed and integrated long-term oceanographic (sea surface temperature, Chlorophyll-*a*, altimetry and upwelling) and fishery data (small pelagic landings) with seabird demographic parameters, evaluating trophic interactions and seabird bycatch to support Marine Stewardship Council-aligned ecosystem approach management.
- Spatial interactions between small pelagic fishing operations and the benthic substrate were evaluated. Spatial modeling enabled the mapping of contact zones between fishing gear and the seabed, as well as the quantification of habitat vulnerability within commercial fishing areas.
- Based on records from onboard observers, the spatiotemporal dynamics of seabird bycatch in small pelagic fishing operations were analyzed. Spatial modeling made it possible to identify specific geographic areas and seasonal peaks associated with higher interaction rates involving protected seabird species.
- Applied quantitative ecological modeling, multivariate time series analysis and remote sensing to evaluate wildlife population dynamics in response to climate oceanographic variability.

## EDUCATION

**Master´s Degree:** Master´s in Science in Recourse Marine Management (2024-2026) in Centro Interdiciplinario de Ciencias Marinas, IPN.

**Bachelor´s Degree:** Biology (2017-2022) in Facultad de Estudios Superiores Iztacala, UNAM.